ZAINAB IRFAN

AI/ML Engineer • Full-Stack Developer • RAG & LLM Systems • AI Automation
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PROFESSIONAL PROFILE

Versatile **AI/ML Engineer** and **Full-Stack Developer** who enjoys turning ideas into reliable, user-friendly products. Experienced in building **RAG systems**, **LLM-powered applications** and **AI automation agents**, along with modern **web applications** using **Next.js**, **React** (frontend) and **FastAPI** (backend), plus **data analytics**, **workflow automation (n8n)** and **REST API integrations**. Comfortable owning a project end to end, from data pipelines and model integration to clean interfaces and cloud deployment, and uses **GitHub Copilot (agent mode)** and the **GitHub Developer Pack** to build efficiently. Collaborative, curious and quick to learn, with experience in remote client work, the Alibaba Cloud AI Hackathon, and 7 industry certifications from AWS, IBM and DeepLearning.AI.

CORE TECHNICAL EXPERTISE

- **AI / LLM Engineering** — RAG Pipeline Architecture · LangChain · Prompt Engineering · LLM APIs (OpenAI, Claude) · LLM Fine-Tuning (LoRA/QLoRA, PEFT) · LLM Evaluation & Benchmarking · Conversational AI Agents · Streaming Chat Interfaces
- **AI Automation & Workflows** — n8n Automation · AI Automation Agents · Workflow Automation · Lead Generation Automation · Cold Outreach Automation · Email Automation · Webhooks & API Integrations · Process Automation
- **Web Development (Full-Stack)** — Next.js · React · Frontend Development · Backend Development · FastAPI · REST API Design · HTML/CSS · JavaScript · Streamlit · Responsive UI
- **Machine Learning & Data Analytics** — Supervised & Unsupervised Learning · Feature Engineering · Model Evaluation · Bias Detection · Exploratory Data Analysis (EDA) · Data Visualization · Predictive Modeling
- **Languages, Libraries & Data** — Python · JavaScript · PyTorch · Transformers · Hugging Face · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn · Qdrant (Vector Database) · Supabase
- **Sports Data & APIs** — BALLDONTLIE API · The Odds API · Odds & Player-Props Data · Third-Party API Integration
- **AI-Assisted Development & Tools** — GitHub Copilot (Agent Mode) · GitHub Developer Pack · Agentic Coding Workflows · Git & GitHub
- **Cloud & Deployment** — AWS (AI/ML) · Render Deployment · Infrastructure Automation · Google Colab (NVIDIA T4 GPU)
- **Professional** — Freelance Client Delivery · Remote Collaboration · Technical Research · Rapid Prototyping · Problem-Solving · Independent Learning
- **Languages Spoken** — English (Proficient) · Urdu (Native)

EXPERIENCE

Freelance AI/ML Engineer — Fiverr (Remote, US Client) | October 2026

Remote contract engagement for a US-based client

- Designed and delivered **NFL Edge**, a custom sports-betting analytics tool built to the client's requirements, owning the work end to end from scoping and data integration to delivery.
- Integrated the **BALLDONTLIE** and **The Odds API** to bring live NFL data and sportsbook odds together in one place.
- Communicated with the client remotely across time zones, translating product needs into working features and iterating on feedback.

Alibaba Cloud AI Hackathon — Participant | 2026

- Collaborated in a team under competitive, time-boxed conditions to design and prototype an AI-powered solution, sharpening rapid prototyping, teamwork and applied problem-solving.

Independent Project Builder | 2024 – Present

- Self-directed development of 8+ full-stack AI/ML projects spanning RAG pipelines, LLM fine-tuning, conversational agents, web applications, data analytics and n8n automations, covering the full lifecycle from API design and data pipelines to model integration and cloud deployment.
- Builds with **GitHub Copilot agent mode** and the **GitHub Developer Pack** to accelerate development, refactoring and code review.

KEY PROJECTS

NFL Edge — Sports Betting Analytics Tool (Fiverr, US Client) | 2026

Stack: Python, BALLDONTLIE API, The Odds API, REST APIs

- Built a custom NFL analytics tool for a US client that surfaces betting **edges** by comparing player props and odds across multiple sportsbooks against live NFL player and game data.
- Integrated the **BALLDONTLIE API** for NFL stats and the **Odds API** for sportsbook lines and props, normalizing both sources into one consistent data layer.
- Delivered as a remote client project on Fiverr, from requirements through final handoff.

Skills Applied: *REST API Integration, Odds & Props Data Modeling, Data Normalization, Sports Analytics, Client Delivery*

Prism AI — RAG-Based Document Intelligence Platform | 2026

Stack: Next.js, FastAPI, LangChain, Qdrant, Supabase, OpenAI/Claude APIs | Deployed on Render

- Built a full-stack RAG platform that ingests multi-format documents (PDF, DOCX, PPT, CSV) and returns structured, context-aware answers.
- Designed the retrieval pipeline with LangChain and Qdrant for semantic search, using Supabase for persistent storage of documents and chat history.
- Integrated OpenAI and Claude APIs with streamed responses for a real-time conversational Q&A experience.
- Built an analytics dashboard with interactive charts for queries per day, question volume and engagement trends, and deployed a production build (Next.js frontend, FastAPI backend) on Render.

Skills Applied: *RAG Pipeline Design, LangChain, Vector Search (Qdrant), FastAPI, Next.js, Streaming Interfaces, Cloud Deployment, Data Visualization*

Green Flora — Smart Agriculture Platform | 2026

Stack: Next.js, FastAPI, LLM APIs, REST APIs | Full-Stack Deployment

- Built a smart agriculture platform that unifies an AI farming assistant, weather insights, an image-based Crop Doctor, live market trends, farm management and a profit calculator in one farmer-friendly dashboard.
- Developed a multilingual assistant (English, Urdu, Roman Urdu) that answers farming questions by text or voice using live weather, market and agricultural data.
- Engineered the Next.js frontend and FastAPI backend, integrating real-time data with LLM-based crop analysis.

Skills Applied: *Next.js, FastAPI, REST API Design, LLM Integration, Multilingual AI, Full-Stack Deployment*

Lead Generation & Cold Outreach Automation | 2025 – 2026

Stack: n8n, Python, LLM APIs, Email Automation, REST APIs

- Built an end-to-end **AI automation** that finds and organizes prospective leads, then runs personalized **cold outreach** campaigns without manual effort.
- Used LLM APIs to write context-aware outreach messages tailored to each lead, and automated sending and follow-ups with **n8n** workflows.
- Connected multiple tools through APIs and webhooks into one repeatable pipeline that saves hours of manual prospecting work.

Skills Applied: *n8n Workflow Automation, AI Automation Agents, Lead Generation, Cold Outreach, LLM Prompt Engineering, API Integration, Python*

Efficient LLM Fine-Tuning with LoRA & QLoRA (TinyLlama-1.1B-Chat) | 2026

Stack: Python, PyTorch, Transformers, PEFT, TRL, BitsAndBytes, Hugging Face Datasets | Google Colab (NVIDIA T4)

- Fine-tuned TinyLlama-1.1B-Chat for Text-to-SQL generation using LoRA and QLoRA on ~3,000 curated examples from the Hugging Face b-mc2/sql-create-context dataset.
- Trained over 3 epochs on an NVIDIA T4 GPU, reaching **84.2% validation accuracy**.
- Applied 4-bit quantization and low-rank adaptation to substantially cut GPU memory use and training cost while preserving model performance.

Skills Applied: *LLM Fine-Tuning, LoRA, QLoRA, PEFT, 4-bit Quantization, GPU Optimization, Hugging Face, PyTorch*

AI Customer Support Agent for Shopify Stores | 2025 – 2026

Stack: Python, LLM APIs, Shopify Integration

- Developed an AI support agent that integrates with a Shopify storefront and handles customer queries automatically.
- Used LLM APIs to generate context-aware responses, reducing manual reply workload for store operations.

Skills Applied: *LLM API Integration, Conversational AI Agents, Third-Party API Integration, Python*

Automated Email Reply System | 2025

Stack: Python, LLM APIs, Email Automation, n8n

- Built an automation that reads incoming emails, drafts context-appropriate replies with an LLM and sends responses automatically.

Skills Applied: *Workflow Automation (n8n), LLM Prompt Engineering, Python, Email/API Integration*

Comprehensive Clinical Data Analysis & Visualization | 2025

Stack: Python, Pandas, Matplotlib/Seaborn, Scikit-learn

- Performed end-to-end EDA on raw clinical datasets, including cleaning, transformation and feature engineering.
- Visualized key trends such as patient distribution by disease, and applied machine learning to generate predictive insights from patient records.

Skills Applied: *EDA, Feature Engineering, Pandas, Data Visualization, Scikit-learn, Predictive Modeling*

CERTIFICATIONS

AI Fundamentals & Cloud (AWS) | **AI Workflow: Feature Engineering & Bias Detection** (IBM) | **Building Systems with the ChatGPT API** (DeepLearning.AI) | **Machine Learning for All** (University of London)

Machine Learning (Simplilearn) | **App Deployment & Security** (Packt) | **Learn to Code with AI** (Scrimba)